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ECG arrhythmias classification based on deep learning methods and transfer learning technique

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ABSTRACT

An intelligence-based electrocardiogram (ECG) signal classification algorithm is very effective in monitoring Cardiac arrhythmias and helps the specialist make a decision and start a safe treatment routine for patients. Preventing wrong decisions about this category of patients that may even lead to death is one of the biggest challenges in the field of medicine, which requires very detailed analyses of the cardiac signals recorded by the patients. The diagnosis accuracy of cardiovascular diseases (CVDs) is largely dependent on the type and space of the extracted features and the ability of the learned classifier. This paper proposes an efficient 34-layer ResNet deep network to classify three types of CVDs based on the features extracted from the time–frequency domain as a scalogram. Also, the introduced ResNet-34 model is combined with the transfer learning technique and the results improvement has been shown. The presented algorithms are compared to other deep networks such as two different structures of a convolutional neural network (CNN) and a recurrent neural network (RNN). Also, these results are evaluated with a dictionary learning-based sparse Non-Negative Matrix Factorization (SNMF) classifier. The results show that the model presented based on ResNet-34 has better performance in different types of evaluation criteria such as accuracy, sensitivity, and robustness that has a positive role in clinical practices.

1. Introduction

Human health is threatened by Cardiac arrhythmias resulting from cardiovascular diseases. These arrhythmias show themself as irregular heartbeat and cause the patient's heart to beat faster, slower, or change to an abnormal form [1]. An electrocardiogram is a non-invasive method to record the fluctuations of the heart's circulatory system which may damage other organs of the patient [2]. Traditionally, diagnosis of heart disease is possible only by observing the regularity of heartbeat patterns and by a specialist doctor but nowadays, using machine learning techniques, high accuracy can be achieved in this medical case. An important issue that increases the accuracy of diagnosis based on machine learning techniques is the extraction of appropriate features from the ECG signals. These features can include temporal features, statistical features, frequency features, and time–frequency features. Also, the learned models that perform classification according to the extracted features can be based on the basic classifiers such as support vector machine (SVM), neural network (NN), and decision tree or other classifiers based on deep learning, which have received a lot of attention in recent years [3–5]. The deep learning-based classifiers such as residual neural network, convolutional neural network, recurrent neural network, and ResNet can detect more types of arrhythmias with higher accuracy [6]. In [7], discrete wavelet transform (DWT) and empirical mode decomposition (EMD) were employed to reduce noise and detect QRS waves in an ECG pulse. Also, the SVM classifier was used to identify the arrhythmia type. An ECG classifier based on higher order spectra and principal component analysis (PCA) for noise reduction along with an SVM classifier was introduced in [8]. An algorithm for the identification of premature ventricular contractions as eight different arrhythmia types based on discrete wavelet transform coefficients and probabilistic neural networks was proposed in [9]. In [10], the combination of classifiers is utilized to increase the diagnostic accuracy of four cardiac arrhythmia types based on SVM. The feature vector in this paper, includes wavelet coefficients, local binary patterns, and wavelength values. In [11], the detection of premature ventricular contractions from different arrhythmias was performed using feature vectors derived from wavelet coefficients in combination with the morphological features. The employed classifiers were neural network, SVM, and K-nearest neighbor (K-NN). An online PCA-based algorithm was introduced to classify arrhythmias in [12]. In this paper, the main abnormal aspects

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and changes in ECG pulses were considered. A PCA-based method to detect normal and abnormal ECG signals based on discrete wavelet conversion coefficients was presented in [13]. In [14], twelve morphological coefficients in combination with Tiger energy operator coefficients and discrete wavelet transform features were utilized to learn a neural network.

The proposed method makes Another type of ECG classification method in comparison with traditional neural networks is based on deep learning structures such as convolutional neural networks (CNNs) that have a remarkable ability in data modeling in the field of signal processing, data retrieval, and ECG arrhythmia classification [15–19]. In these algorithms, a neural network with deep structure is designed to precisely represent the ECG signal content. Compared to basic machine learning-based classification methods such as statistical-based, clustering-based, and SVM-based, deep learning-based ECG classification algorithms could properly map the content of ECG signals to their corresponding class resulting from the powerful capability of multi-layers. In [20], an end-to-end CNN model was employed with an automatic feature extraction step from noisy signals to perform ECG classification. In [6], a 1-D CNN network was proposed to diagnose five types of heartbeat features based on the wavelet denoising method. [21] reviews the deep learning techniques, preprocessing, methodology, and evaluation procedures that have been extensively introduced for ECG arrhythmia diagnosis. A hybrid 2-D CNN combined with the long short-term memory (LSTM) network was proposed in [22]. Also, this approach uses continuous wavelet transform to transform a 1-D ECG signal to a 2-D scalogram-colored image to be injected into the deep network. In [17], a pre-trained densely connected convolutional neural network (DenseNet) as a feature extractor and SVM classifier was proposed to detect four arrhythmias using spectrograms of the transformed ECG signals. A CNN-based algorithm was utilized to diagnose seven premature ventricular contractions [18]. The input of this intelligent system is the raw time series of ECG signals and the arrhythmia types are detected from the output softmax layer. A combination of convolutional autoencoders (CAEs) and transfer learning (TL) was designed to classify normal and abnormal states of ECG signals using their spectrogram mapping [19]. In this paper, high-performance deep features extracted from a Resnet34 network learned by a 2-D domain version of ECG signals are employed to classify three arrhythmia types of CVDs.

significant advancements in the medical field, including the following contributions:

- Novel ECG Arrhythmias Classification technique: The introduction of an innovative method for detecting ECG Arrhythmias helps to mitigate the issue of imbalanced ECG data, which can improve the accuracy and effectiveness of classification models.
- Effective solution: The proposed approach effectively addresses the challenges associated with ECG Arrhythmias Classification, including imbalanced data, representation, and model complexity. By using novel deep learning models with the multi-attention mechanism and transfer learning to enhance the performance of classifiers.
- Optimized deep ResNet models: The developed and optimized deep ResNet models, which were trained using annotated data, can effectively classify ECG Arrhythmias categories under various scenarios, making them adaptable and efficient.
- Transfer learning: By leveraging the benefits of transfer learning, the proposed system can design a fine-tuned ResNet model, which can efficiently enhance the performance of the ECG Arrhythmias Classification task.
- Scalable and effective solution: The presented algorithm can provide an effective and scalable solution for enhancing ECG Arrhythmias Classification performance, with potential applications in medical applications.

Section 2 includes a detailed description of the proposed approach including the database, pre-processing steps, the features mapping and conversion of data, and parameter setting for the designed architecture. Experimental results and performance evaluation are reported in Section 3. In Section 4, the results of using the proposed method are discussed and evaluated and in the last Section, the conclusion is made.

2. ECG data processing steps

ECG signals processing is a test way to monitor heart health from electrical activity recorded from heart rhythm. Due to the importance and prevalence of different types of heart diseases, it is necessary to design computer-aided tools to measure human heart health with the highest accuracy. Also, the accurate diagnosis of cardiovascular disease depends on specialist experience. In recent years, state-of-the-art performances for ECG classifiers achieved by deep learning-based networks. In this paper, a ResNet-based ECG classifier is designed to distinguish three ECG arrhythmia types. Different materials and information to run this architecture in the presented algorithm are listed in the following.

2.1. Dataset

The PhysioNet arrhythmia dataset is used to evaluate the performance of the proposed arrhythmia diagnosis method, which includes the labels of various arrhythmias for categorized cardiac signals [23]. This study leverages 162 ECG recordings obtained from three prominent PhysioNet databases: the MIT-BIH Arrhythmia Database, the MIT-BIH Normal Sinus Rhythm Database, and the BIDMC Congestive Heart Failure Database with various age ranges [23–25]. Each patient recording was composed of 65,536 sequence segments. A normal heartbeat with waves is shown in Fig. 1.

The recorded cardiac signals are publicly available and related to 22 women in the age range of 23–89 and 25 men in the age range of 22–79 years. The central objective of this investigation is to develop a classifier capable of accurately distinguishing between arrhythmia (ARR), congestive heart failure (CHF), and normal sinus rhythm (NSR). 96 recordings of train data were selected from the MIT-BIH cardiac arrhythmia database with a sampling rate of 128, sampling frequency of 360 Hz, and 5 μV/bit [24]. 30 recordings were taken from the MIT-BIH normal sinus rhythm database with a sampling frequency of 128 Hz and 12 bits [25]. Also, 36 recordings from patients with normal sinus rhythms with a sampling frequency of 128 Hz. Two signals were used to record each ECG signal: the modified limb lead II and one of the modified leads V1, V2, V4, and V5. To ensure proportional representation, data augmentation techniques have been employed, resulting in 30 recordings each from the NSR, CHF, and ARR categories. For further insights, Table 1 provides a comprehensive summary of the relevant information embedded within all arrhythmia-related ECG signals sourced from PhysioNet. Table 1 summarizes the characteristics of the arrhythmia databases available on PhysioNet.

2.2. Pre-processing

Pre-processing is an important step in a deep learning-based ECG classification algorithm for many of the same reasons as in a traditional ECG classification algorithm. Here are some additional reasons why pre-

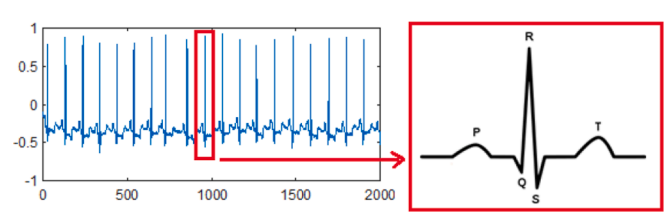

Fig. 1. A normal heartbeat signal with its constituent waves.

Table 1
The summarized characteristics of employed PhysioNet arrhythmia database.

Arrythmia class	No of recording	No of augmented Recording	No of Samples	Sampling Rate (HZ)
ARR	96	38	38 × 65,536	128
CHF	30	38	38 × 65,536	360
NSR	36	38	38 × 65,536	128

processing is important in deep learning-based ECG classification such as, data preparation that denotes deep learning algorithms require a large amount of labeled data to train the model. Pre-processing steps are employed to properly prepare the data for the deep learning model, such as removing noise, normalizing the data, and splitting the data into training and testing sets. Feature extraction step that mentions deep learning algorithms need to extract relevant features from the ECG data. Pre-processing can help to extract these features, such as heart rate, heart rate variability, and other features that the deep learning model can use to distinguish between different types of cardiovascular disease. In data augmentation step, deep learning algorithms can benefit from data augmentation or the process of generating new data by altering the original data. Pre-processing can also help to optimize the performance of the deep learning model by reducing overfitting and improving the generalization ability of the model in the model optimization step. An ECG signal is prepared after pre-processing for training a statistical or deep model performed usually in different steps as follows.

2.2.1. Data denoising

ECG signals in a therapeutic environment are likely to be recorded with different types of interferences such as power frequency noises, muscle movement, power-line interface (PLI), and Electromyography (EMG) signals. These low or high-frequency signals must be removed from the ECG signal before the feature extraction or training step. Different categories can be employed to denoise the recorded ECG signals: filter-based denoising method as a traditional approach [26,27], wavelet-based-denoising methods [28,29,6], and hybrid denoising algorithms with a combination of the results of different filters [30,31]. The basic filter-based denoising algorithms due to easier implementation and high effectiveness are more popular than the two other categories. One of the types of noise signals involved in the cardiac signal is the power-line interface noise signal, which may be caused by damage to the electrodes when the signal is recorded. In this paper, the effect of this noise is eliminated by centering the cardiac signal, which removes the average value from the ECG signal. To achieve the mentioned features, the heart rate pulses must be accessed. Also, an adaptive median filter is utilized to smooth the raw ECG data [32].

2.2.2. Data segmentation

Deep learning methods for feature extraction need a large amount of data for training. In the PhysioNet arrhythmia datasets, each heartbeat is labeled with an ECG arrhythmia. In this paper three ECG arrhythmias are considered to classify: cardiac arrhythmia (ARR), congestive heart failure (CHF), and normal sinus rhythm (NSR). Each ECG signal of every class is segmented into 360 samples as mentioned before and centered around the R-peak in the heartbeat [33].

To solve the problem of capturing multiple heartbeats within a single segment, one common strategy is to use a sliding window approach that is employed in this paper. The ECG signal is divided into overlapping segments, each corresponding to a single heartbeat. For each segment, a scalogram is generated, resulting in an image representation. These individual heartbeat images are then fed into the deep model for classification. The model learns to recognize patterns specific to each type of arrhythmia based on these segmented images.

2.2.3. Data conversion and mapping

Most learning-based ECG classifiers use 1-D cardiac signals which usually have different noises or artifacts. Using a 1-D raw time series of ECG Signals may result in the need for many complicated pre-processing steps. In these cases, the ECG signals are transformed into a 2-D image using a mapping transform [17,22]. This conversion maps the ECG signal into a 2-D colored scalogram using the continuous wavelet transform (CWT) with a resolution of 227 × 227 × 3. After this transformation, the frequency content of the ECG signal in a time-scale domain can be visible. The ARR, CHF, and NSR ECG signals and their scalograms based on the wavelet transform are shown in Figs. 2-3.

2.2.4. Data augmentation

Training a deep network with the additional data can usually reduce the overfitting rate and cause an effective ability for machine learning models [34–36]. On the other hand, a deep model needs massive input data for learning, and therefore enlarging the input data through different approaches is a necessary problem. Also, the problem of limited data can be solved easily by artificially expanding the dataset and adding it to the training set. The new images can be resulted from, rotating or adding some noise signals. In clinical data analysis usually, recording abnormal classes is more difficult than normal cases and as a result, the number of these data is much less than others [21,37]. On the other hand, if a deep model is learned with an unbalanced number of different categories, then overfitting or biased learning may occur. Therefore, in this paper, the data augmentation technique is used to obtain more data for the categories that have fewer training samples [38]. The perturbation-based method is used for data augmentation to add extra data to the ECG database by adjusting initial data samples in the way of scaling or shifting to enlarge the dataset artificially [39]. Some augmented ECG signals from three categories of mentioned arrhythmias captured by data augmentation algorithms defined in [40] are shown in Fig. 4.

2.3. Methodology and architecture

In this section, the proposed algorithm is presented using the model learning of the extracted scalogram of the ECG signals. A Resnet34 architecture is employed to represent the proper structure of the input scalogram image and classify three arrhythmia types of CVDs, ARR, CHF, and NSR [41,42]. To select a proper deep model to solve this classification task, various deep learning models were explored and ultimately chosen ResNet34 due to its effectiveness in handling ECG signal features, its ability to address the vanishing gradient problem through

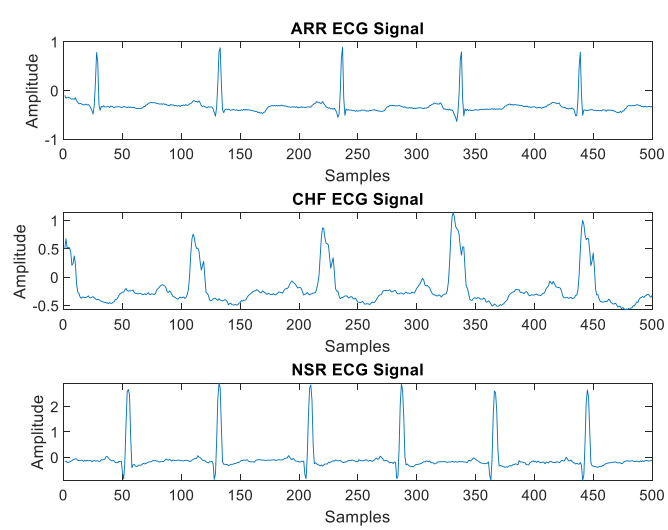

Fig. 2. Distribution of ECG waveforms from three classes of arrhythmias: ARR, CHF, and NSR.

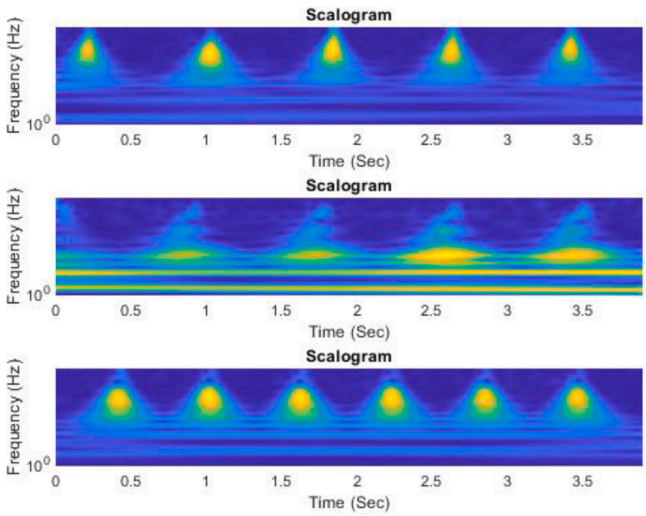

Fig. 3. Representations of 227 × 227 × 3 Scalogram of ARR, CHF, and, NSR ECG signals of Fig. 1 using CWT.

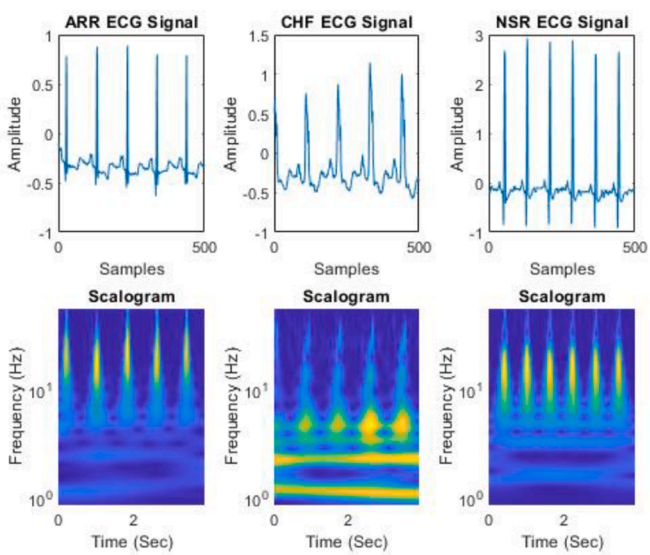

Fig. 4. Some examples of data augmentation of the ECG dataset for ARR, CHF, and NSR categories.

skip connections, and its balance between model complexity and performance. It solves the known vanishing gradient problem because the network is too deep and the loss function is calculated to easily shrink to zero after several runs of the chain rule. ResNet34 offered a desirable balance between model complexity and performance. Additionally, ResNet34′s pre-trained weights from large image datasets make it suitable for transfer learning in this task, reducing training time and improving generalizability to perform well on unseen ECG data.

2.3.1. *Transfer learning*

Transfer learning is a machine learning technique that involves reusing features or knowledge from a pre-trained model to enhance the performance of a new task or domain, rather than training the model from scratch [43,44]. It is often used to improve the efficiency and accuracy of deep learning models on new tasks or domains, especially when there is limited or no labeled data available for the new task. In this regard, transfer learning can be viewed as a type of knowledge transfer, where the knowledge gained from one domain is transferred to another. The applications of transfer learning are numerous, particularly in the fields of natural language processing, computer vision, signal processing, and different classification tasks. Also, transfer learning is a powerful technique for improving the performance of ECG arrhythmias classification, especially when labeled data is limited or expensive to obtain. Transfer learning can help reduce the time and resources required to acquire and label a new dataset, which can be challenging for small datasets or datasets with limited availability. One of the main applications of transfer learning for ECG arrhythmias classification is in the area of unbalanced and imbalanced datasets. By using a pre-trained model on a large, labeled dataset, the imbalance in the data can be reduced. Additionally, using a pre-trained model can also help to generalize new datasets and tasks, reducing the need for new training data. Another key application of transfer learning for ECG arrhythmias classification is in the area of feature engineering. By using a pre-trained model on a large, labeled dataset, the feature representation can be optimized, reducing the dimensionality and reducing the computational cost of the model. Furthermore, transfer learning can also be used to improve the performance of ECG arrhythmias classification models by fine-tuning a pre-trained model on a smaller, labeled dataset, reducing the need for labeling new data sets which can be time-consuming and costly. It is worth noting that transfer learning for ECG arrhythmias classification has several potential limitations, such as the need for carefully selecting the pre-trained model and the need for carefully designing the transfer learning process to ensure that the pre-trained model generalizes well to the new dataset and task. It is important to carefully consider the benefits and limitations of transfer learning when developing an ECG arrhythmias classification model [45,46]. For ECG classification, ResNet34 can be used as a pre-trained model and the output features from a specific layer in the network can be used as inputs to a classifier. Using an ImageNet pre-trained model, such as ResNet34, for ECG classification is a common approach in deep learning-based ECG classification. ImageNet is a large-scale visual recognition dataset and has been used to pre-train deep neural networks for image classification tasks. The features extracted by ResNet34 can be used to represent ECG data in a way that is suitable for ECG classification tasks. The features extracted from each layer of the network capture different levels of abstraction and can be used to learn complex, nonlinear patterns in the data. By using ResNet34 as a pre-trained model, the need for labeled ECG data for training can be reduced, as the network can learn from the ImageNet data. In this paper, ResNet34 has been pre-trained on a large-scale dataset, which allows it to generalize well to new data and effectively learn from ECG data. ResNet34 is also able to learn features that correspond to the spatial structure of ECG signals, making it well-suited for ECG classification. Using ResNet34 as a pre-trained model can lead to significant improvements in ECG classification performance compared to traditional machine learning techniques. It can also reduce the need for labeled ECG data for training, which can be time-consuming and expensive to obtain. However, it is important to carefully consider the specific features extracted by each layer of the network and the type of classifier used, as these factors can impact the performance of the ECG classification model [43–46]. The detailed breakdown of the proposed ResNet34 when combined with transfer learning is described in the following. A pre-trained ResNet34 model comes into play. This model has already been trained on a massive image dataset like ImageNet, allowing it to learn low-level features like edges, shapes, and basic textures. In transfer learning, pre-trained ResNet34 is used for feature extraction. These layers excel at extracting generic visual features that can be repurposed for ECG data, even though it's not an image dataset. Early layers in CNNs like ResNet34 tend to capture generic features that are applicable across different visual domains. ECG data, although not an image, still has underlying patterns and structures that these layers can effectively learn. In this manuscript, the weights of the pre-trained convolutional layers are partially adjusted during training for ECG classification. This allows the model to adapt the generic features learned on ImageNet to the specific patterns in ECG data. Fine-tuning is often used with larger ECG datasets where there's more data for the

model to learn from. Also, the new layers were added on top of the pre-trained convolutional layers. These new layers are typically fully connected layers specific to the ECG arrhythmia classification task. These layers learn higher-level features specific to ECG patterns and perform the final classification of different arrhythmia types [45,46].

2.3.2. Deep model structure

With ResNets, the gradients can flow directly through the shortcut connections and ignore one or more layers [41,42].

The structure of the primitive block of a ResNet-34 model used for all 34 layers is based on a residual building block composed of several convolutional layers (Conv), rectified linear units (ReLU) activation functions (AF), and max-pooling followed by a fully connected layer (FCNN). Also, a ResNet uses different learning techniques such as batch normalizations (BN). The convolution layers convolve the independent features of the previous layer using coevolution kernels and map them into a nonlinear activation function as

$$y_{l,k}^{i} = f(b_{l,k}^{i} + \sum_{n=1}^{N} \omega_{l,k}^{i,n} * x_{l,k}^{i,n} \tag{1}$$

where $y_{l,k}^{i}$ is the output of the ith neuron in lth layers and kth convolution kernel. Also, $f()$ is the activation function, $x_{l,k}^{i,n}$, $\omega_{l,k}^{i,n}$, and $b_{l,k}^{i}$ are input data, the convolutional kernel coefficients, and the bias value in nth dimension, respectively [21].

Each convolutional layer is followed by a batch normalization layer and a ReLU activation function. A ReLU is a non-linear activation function that returns zero if it receives any less than zero input and deactivates the neurons, but for any positive input value, it returns that value.

$$\mathrm{ReLU}\left(y_{l,k}^{i}\right) = \max(0, y_{l,k}^{i} \tag{2}$$

The pooling layers reduce the complexity of the deep model with the dimension reduction of the convolution layer outputs to avoid the overfitting problem. These layers' averages or the other versions maximize the convolution layer outputs using the pooling function [22].

$$z_{l,k}^{i} = f\left(\alpha_{l,k}^{i} pool\left(x_{l,k}^{i,n}\right) + b_{l,k}^{i}\right) \tag{3}$$

where $z_{l,k}^{i}$ is the output of the ith neuron in lth layers and kth convolution kernel. $f()$ is the activation function in the pooling layer, *pool()* is the averaging or maximization pooling function, and $\alpha_{l,k}^{i}$ is the weight coefficients matrix. Different blocks of a deep network usually begin with convolution and pooling layers to map the image into the structured features and analyze them. In the next step, the result of this process feeds into a fully connected layer to make the final classification decision using a linear transformation of the input vector with a weight matrix.

$$z_{l,k}^{i} = f\left(\omega_{l,k}^{i,n} * x_{l,k}^{i,n} + b_{l,k}^{i}\right) \tag{4}$$

where $z_{l,k}^{i}$ is the output of the ith neuron in lth layers and kth convolution kernel. $f()$ is the activation function in the last layer. Also, the BN block normalizes the input of activation functions and scales them to speed up the learning process regularly and enable the network to be learned with high performance. This technique is done using normalization of each layer's inputs, the mean and variance values, usually set to zero mean and unit variance with δ and β parameters.

$$\delta = \frac{1}{n}\sum_{i=1}^{n} z^{i},\ \beta^{2} = \frac{1}{n}\sum_{i=1}^{n} z^{i} - \delta,\ \delta\widehat{z^{i}} + \beta = BN(z^{i}) = \frac{z^{i} - \delta}{\sqrt{\beta^{2} + \varepsilon}} \tag{5}$$

where $\widehat{z^{i}}$ is the normalized version of z^{i} and ε is set to 10^{-2}.

The ResNet-34 model in the proposed ECG classifier algorithm with all designed layers is shown in Fig. 5. It should be noted that a deeper residual network such as ResNet-50 does not guarantee better results for a classification goal and the results may be even lower than the traditional machine learning-based methods because model over-fitting may be occurred. So, after some experimental simulations, ResNet-34 was selected in this paper the trade-off between the input data dimension and the model complexity.

The adaptive moment estimation (Adam) optimizer with an initial learning rate of 10^{-3} is used to learn ResNet-34 which is multiplied by 0.1 at epochs 40 and 80 [47]. This optimizer is an optimization algorithm for stochastic gradient descent (SGD) to train deep learning model layers. Also, the learning rate is decreased every 15 epochs by 50 %. For efficiency, the number of iterations is set to 10, and train the ResNet-34 for 50 epochs at each iteration. In all other experiments, the network is trained for 200 epochs till convergence. The batch size according to the experimental results is set to 128 since increasing the batch size often leads to specific loss and lower test accuracy. These network hyper-parameters are summarized in Table 1.

The proposed algorithm was trained on a PC with an Intel Xeon E5 2600 CPU (16-cores), 3.20 Ghz of RAM, and a Linux machine running Ubuntu 18.04 with 192 GB of DDR4 RAM.

The compared methods are different models of CNN presented in [6,18,22], and a recurrent neural network (RNN) in addition to the model learning-based ECG classifier method using sparse Non-Negative Matrix Factorization (SNMF) presented in [48]. In [6], a 1-D CNN network with raw ECG data was proposed to categorize five types of arrhythmias using the wavelet denoising method. A hybrid of 2-D CNN and LSTM networks was introduced in [22]. A continuous wavelet transform analysis was utilized to convert a 1-D ECG signal to a 2-D scalogram image and learn a CNN network. In [18], a CNN-based algorithm was used to classify seven CVDs. In this algorithm, features are learned from the raw time domain ECG signals and manually replaced with the extracted features to help the Cardiologists in screening the patient's illness. In [48], the morphological features in combination with the coefficients extracted from different decomposition levels of wavelet packet transform were employed to dimensionality reduced by the sparse structured principal component analysis and modeled with the sparse non-negative matrix factorization algorithm. It has detected three types of arrhythmias along with the normal heart rate with a high accuracy.

In this paper, the 1D CNN was run with a batch size of 36, a learning rate of 10–3, and a hidden node of 20 [6]. Using CNN, 30,000 learning scenarios were run to learn the dataset, 5000 for the validation set, and 10,000 for the test dataset.

In the simulations, 30,000 learning scenarios choose to balances the

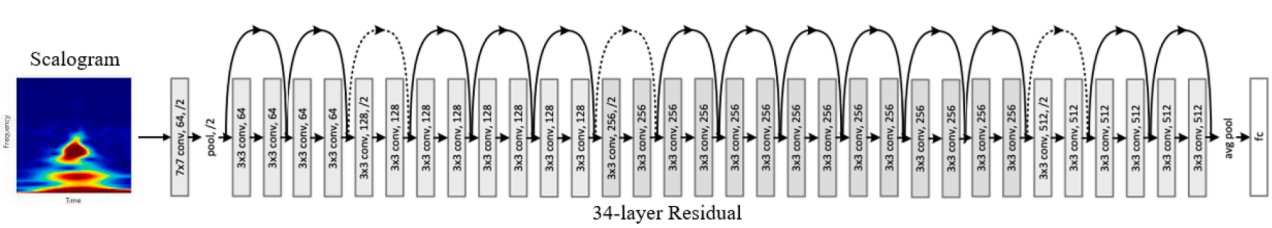

Fig. 5. The different layers of the ResNet-34 model structured in the proposed method.

need for enough data to train the model on complex ECG patterns for arrhythmia detection with potential limitations in data availability and computational resources. It allows for a diverse dataset to train the model effectively and avoid overfitting, while acknowledging that even more data or data augmentation techniques could be beneficial depending on the model's complexity. Overall, 30,000 scenarios were strategically selected to leverage transfer learning and maximize the effectiveness of deep learning for ECG arrhythmia classification.

The optimal hyperparameters for the 2D CNN model include learning rate 10–2, hidden units 32, batch size 10, and epochs 20 [22]. These parameters related to all deep models utilized in this paper can be seen in Table 2. These models involve different convolutional layers, pooling layers, and one full-connected layer. The convolutional layers are followed by pooling parts which are used to downsample the output feature maps to increase the robustness of the model and avoid over-fitting. For the ResNet34 model, a 3 × 3 kernel size for all 34 convolutional layers and max-pooling was considered. The architecture of the designed ResNet-34 is shown in Table 3. The model in the CNN was learned based on the Adam optimizer solver. The learning rate, weight decay, and batch size were set to 10–4, 2 × 10–4, and 64, respectively. The epoch number is 100∼200. The softmax activation function was used with 128 nodes per layer. In the final layer, the CNN model calculates the probability that the input frame is normal or abnormal. A threshold assessment method was determined as the class of the input frame. An overview of the CNN architecture for voice activity detection can be seen in Fig. 6 which contains different layers such as a convolutional layer to map features, pooling, fully connected, and softmax. An RNN network with feedback and softmax layers is depicted in Fig. 7. The output of the recurrent hidden layer, with 400 units, is temporally mean-pooled and followed by a fully connected output layer with softmax activation and cross-entropy loss over the frame class.

The block diagram of the proposed method is shown in Fig. 8. Fig. 9 reports the training progress of the presented ResNet-34 model learned based on the parameters and architecture defined in Tables 1-3 according to the accuracy and loss function in the train and test steps.

To evaluate the performance of an ECG classifier, different measures are used such as accuracy (Acc) Specificity (Spe), Positive prediction rate (Ppr), Sensitivity (Sen) or Recall (R), F-Measure, and confusion matrix. The accuracy rate shows the ability of the classifier to detect the true class of the input frame. The specificity index represents the ability to classify an ECG signal negatively. The Ppr measure reports the positive proportion of the correct labeled frames, and Sen calculated the ability to distinguish different arrhythmias.

The measures employed in this paper to compare different models are mentioned in the following:

$$\mathrm{Acc} = \frac{TP + TN}{TP + TN + FP + FN} \tag{6}$$

$$\mathrm{Spe} = \frac{TN}{TN + FP} \tag{7}$$

$$\mathrm{Ppr} = \frac{TP}{TP + FP} \tag{8}$$

Table 3
The employed ResNet-34 architecture.

Layer name	Data output size	Layers
Conv1	112 × 112	7 × 7,64, stride 2
pool1		3 × 3, max pool, stride 2
Conv2_x	56 × 56	$\begin{bmatrix} 3\times 3, 64 \\ 3\times 3, 64 \end{bmatrix} \times 3$
Conv3_x	28 × 28	$\begin{bmatrix} 3\times 3, 128 \\ 3\times 3, 128 \end{bmatrix} \times 4$
Conv4_x	14 × 14	$\begin{bmatrix} 3\times 3, 256 \\ 3\times 3, 256 \end{bmatrix} \times 6$
Conv5_x	7 × 7	$\begin{bmatrix} 3\times 3, 512 \\ 3\times 3, 512 \end{bmatrix} \times 3$
fc1		1000-d fc, softmax
pool time		average pool, stride 1

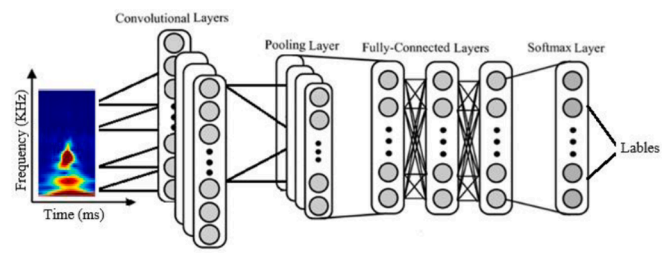

Fig. 6. The CNN architecture used in the arrythmia classification algorithm.

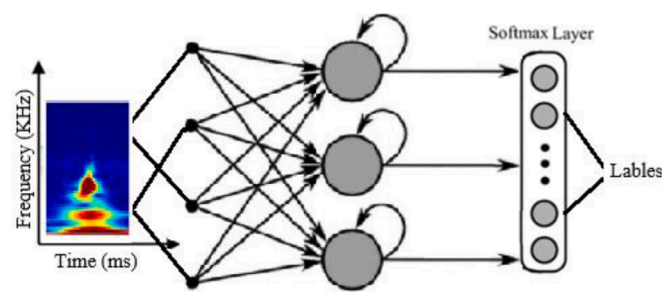

Fig. 7. The RNN architecture employed in the arrythmia classification algorithm.

$$\mathrm{Sen}\,(\mathrm{R}) = \frac{TP}{TP + FN} \tag{9}$$

$$\mathrm{F-Measure} = 2 \times \frac{\mathrm{Ppr} \times \mathrm{Sen}}{\mathrm{Ppr} + \mathrm{Sen}} \tag{10}$$

where TP, TN, FP, and FN introduce True Positive, True Negative, False Positive, and False Negative, respectively [6]. TP stands for the number of correct detected arrhythmias classes. TN is the number of negative frames that are correctly classified as negative. FP shows the frames that are wrongly predicted as positive and FN stands for the number of

Table 2
The hyperparameters set for the different used deep networks.

	Epochs	Optimizer	Batch Size	Kernel size	Activation function	Learning rate	Weight decay	#of parameters	# of total layers
1-D CNN [P1, P5]	60	Adam	36	3 × 3	Sigmoid	10^{-3}	10^{-4}	∼	20
2-D CNN [P3]	20	Adam	10	3 × 3	Sigmoid	0.01	10^{-4}	∼	32
RNN	200	SGD	64	3 × 3	Sigmoid	10^{-4}	10^{-4}	∼	20
ResNet-34	250	Adam	128	3 × 3	Non-Linear	10^{-4}	10^{-4}	0.40 M	34

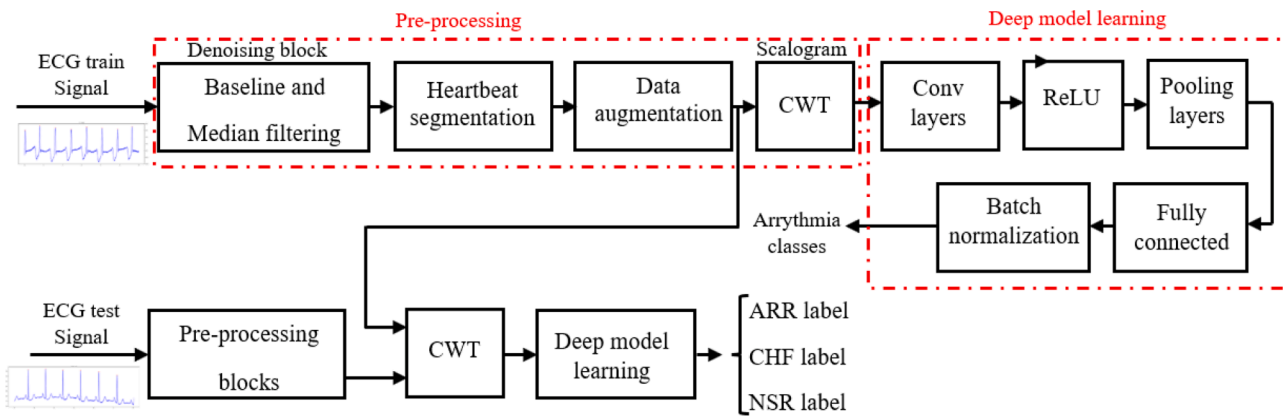

Fig. 8. The block diagram of the proposed deep learning-based ECG classifier.

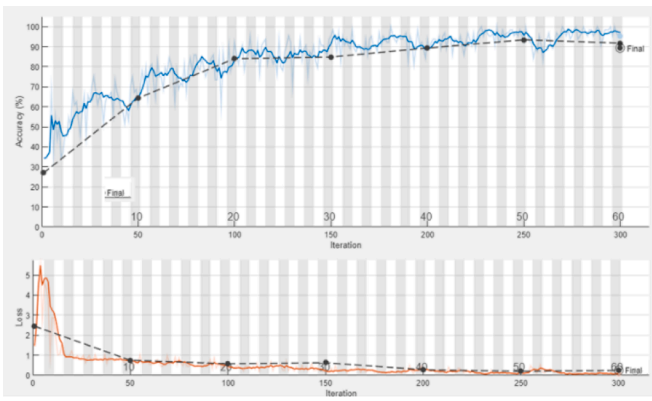

Fig. 9. The proposed ResNet-34 learning procedure in the training step.

positive frames that are incorrectly detected as negative. The larger values for these metrics result in better classification rates.

3. Results and discussion

In this paper, an intelligent system to classify ECG arrhythmia is proposed. The raw time series data after denoising and segmentation is converted to 2D scalogram images to feed the ResNet34 deep model. The input layer receives scalogram sequence images with size 227 × 227 × 3. The output class is captured from the fully connected layer after the batch normalization step. The presented system for classifying three types of ECG arrhythmias is compared with some mentioned methods by varying different parameters such as activation function, Epochs, batch size, and learning rate. The proposed algorithm compared with different models of CNN introduced in [6,22,18], an RNN architecture, and an SNMF-based ECG classifier presented in [48].

3.1. Simulation results using deep model learning

The results of the classification of the ARR, CHF, and NSR ECG signals based on ResNet34 model learning, CNN models, RNN architecture, and SNMF-based classifier are shown in Table 4. The bias and weight parameters in the RNN are updated on a scaled conjugate gradient algorithm. The hyperbolic tangent sigmoid is selected as the hidden layer transfer function and the number of neurons for each hidden layer is set to 30. In the CNN deep models, 50 % dropout regularization for solving over-fitting problems during the training step and ignoring some function nods was regarded.

In this experiment, to learn the deep model in the learning step, the database is divided using 10-fold cross-validation. Then all database is separated equally into 10 sections and in each iteration, 9 out of 10 categories are utilized to learn the model, and the remained section is used for validation. This procedure is iterated by 10 times by moving the training or test sections. The final accuracy rate is calculated by averaging ten yielded measure values. To have a more precise consideration of the performance of the presented algorithm, the overall accuracy, Specificity, Positive prediction rate, Sensitivity, F-measure, and confusion matrix are referenced in Tables 5-6. For a more precise comparison of the performance of the presented algorithms, a statistical test was employed to assess the efficiency between different methods and specify whether there is any statistically significant difference between them. A statistical test such as Friedman statistics is based on a null hypothesis that states all methods compared have the same efficiency [49]. This test determines whether this hypothesis is rejected or accepted and, in the

Table 4
The Accuracy (Acc) of ECG arrhythmia classification for the proposed method in comparison with other mentioned methods.

	ARR	CHF	NSR	Mean
[6]	97.14	97.96	97.58	97.56
[22]	97.34	97.21	97.98	97.51
[18]	97.96	97.85	97.90	97.90
RNN	96.97	96.73	96.02	96.57
SNMF-based [44]	95.73	95.09	96.14	95.65
ResNet34-based (Proposed)	**98.81**	**98.59**	**98.01**	**98.47**

Table 5
The overall Accuracy overall accuracy (Acc), Specificity (Spe), Positive prediction rate (Ppr), Sensitivity (Sen), and F-Measure of ECG arrhythmia classification for the RenNet34-based proposed method.

		Acc (%)	Spe (%)	Ppr (%)	Sen (%)	F-Measure (%)
	ARR	97.14	97.24	96.14	96.36	96.25
[6]	CHF	97.96	97.15	96.71	95.12	95.91
	NSR	97.58	97.99	95.80	95.70	95.75
	ARR	97.34	97.11	96.33	96.08	96.20
[22]	CHF	97.21	97.16	96.54	97.00	96.77
	NSR	97.98	97.42	95.76	96.19	95.97
	ARR	97.96	98.04	96.57	97.15	96.86
[18]	CHF	97.85	97.69	97.01	97.26	97.13
	NSR	97.90	98.10	95.99	96.73	96.36
	ARR	96.97	96.17	96.35	97.11	96.73
RNN	CHF	96.73	96.51	96.14	96.83	96.48
	NSR	96.02	96.33	95.62	96.31	95.96
	ARR	95.73	93.56	95.66	95.94	95.80
SNMF-based	CHF	95.09	93.64	96.41	96.48	96.44
[44]	NSR	96.14	94.05	96.62	96.55	96.58
ResNet34-based	ARR	**98.81**	100	97.51	97.21	97.36
(Proposed)	CHF	**98.59**	98.13	97.63	98.42	98.02
	NSR	**98.01**	98.28	97.62	98.85	98.23

Table 6
The confusion matrix for ECG arrhythmia classification of the RenNet34-based proposed method.

ARR (class 1)	**98.81 %**	0.46 %	0.73 %
CHF (class 2)	0.81 %	**98.59 %**	0.6 %
NSR (class 3)	1.06 %	0.93 %	**98.01 %**
	ARR (class 1)	CHF (class 2)	NSR (class 3)

results, whether the proposed method has better performance or not. The ρ-value as a main parameter in this test has been set to 0.05 in the determined statistical significance level [50]. The ρ-value is the probability of obtaining a result equal to or more extreme than what occurred under the assumption of no effect, no difference, or null hypothesis. This parameter measures how the probability it the difference between classes can be observed [50]. In a statistical Friedman significance test, if the p-value is calculated smaller, there exists stronger evidence to reject the null hypothesis. The ρ −value in addition to the average accuracy of each class in the mentioned methods can be seen in Table 7. In this Table, the null hypothesis is that all methods in the rows of this Table along with the ResNet34-based introduced algorithms perform equally well in the specified significance level. The p-value obtained from Table 7 shows that all ρ-values are lower than the specified level of 0.05, therefore with full evidence the null hypothesis is rejected. Thus, the proposed algorithm works more effectively than the other methods as listed in rows of Table 7 in this classification field. The results show that the proposed method using ResNet-based model learning can classify ECG arrhythmias with high accuracy compared to other methods mentioned in this processing field. Also, the presented algorithm has the same stable results for other metrics such as specificity, prediction rate, sensitivity, and F-Measure over the selected databases.

3.2. Simulation results using transfer learning

In this Section, a new ECG classifier using ResNet-34 architecture which combines with transfer learning is presented. As mentioned before a deep network needs a large number of training data to result in a high classification performance. A main problem in this regard is that it is usually very difficult to have a large available quantity of training data in practice. In the case of cardiac arrhythmias, it is also difficult to access patients with different kinds of diseases, and on the other hand, the labeling of captured ECG signals is very expensive. Therefore, the transfer learning technique is often employed when encountering an insufficient amount of training data [51,52]. In recent research, it has been proven that this technique is a very effective complementary learning algorithm [53,19]. Due to a fixed network structure, the RestNet34 deep model is pre-trained on the ImageNet dataset which contains 100,000 + images across 200 different classes. ImageNet was pre-trained on a large-scale mature dataset. This procedure is performed before training on the PhysioNet arrhythmia datasets. There is often a prominent problem with low data as overfitting because the amount of data in the experiments is relatively small and the model also includes more epochs during the training step. Therefore, a poor model with a lower prediction can be yielded.

Table 7
The statistical test analysis of the different mentioned methods along with the RenNet34-based proposed algorithm.

		Acc (%)	ρ-value	Mean ρ-value	Mean Acc (%)
	ARR	97.14	0.003	0.004	97.56
[6]	CHF	97.96	0.002		
	NSR	97.58	0.008		
	ARR	97.34	0.004	0.003	97.51
[22]	CHF	97.21	0.002		
	NSR	97.98	0.005		
	ARR	97.96	0.006	0.005	97.90
[18]	CHF	97.85	0.008		
	NSR	97.90	0.001		
	ARR	96.97	0.009		
RNN	CHF	96.73	0.011	0.009	96.57
	NSR	96.02	0.008		
	ARR	95.73	0.027		
SNMF-based [44]	CHF	95.09	0.018	0.017	96.65
	NSR	96.14	0.007		
ResNet34-based	ARR	**98.81**	**0.002**		**98.47**
(Proposed)	CHF	**98.59**	**0.006**	**0.003**	
	NSR	**98.01**	**0.001**		

Improved recognition results can be expected through the application of the transfer learning technique and optimization of the ECG arrhythmias classification model. The optimal parameters set for ResNet in combination with transfer learning are according to Table 8 and the architecture is similar to that reported in Table 3. To have a fair comparison and since deep models are very sensitive to parameter setting, the transfer learning technique has only been applied to the proposed structure. Table 9 shows the obtained prediction results for ECG arrhythmias classification of the RenNet34-based proposed method in combination with transfer learning. The confusion matrix obtained by the presented model with transfer learning is demonstrated in Table 10. The reported results show that the designed deep model is well-tuned to represent the different data classes. Also, a lot of learning computation time was saved. The results showed that architecture with transfer learning over all data classes results in accuracy with a 0.3 % improvement in comparison with the designed model without this technique. Table 11 shows the ρ-value of the statistical test analysis of the different mentioned methods and the RenNet34-based proposed algorithm employing the transfer learning technique.

The null hypothesis in this Table is that all classifiers listed in the rows of Table 4 in contrast to the proposed method perform the same and are equally good. Similar to the previous scenario in Section 3.1, the significance level is set to 0.05. The calculated ρ-values are still small and lower than the significant level for each three classes.

Therefore, it results in that the null hypothesis should be rejected and the proposed deep model performs much better than the rest of the classifiers. The training progress procedure included the accuracy and loss function in the train and test steps of the presented ResNet-34 model learned based on the transfer learning technique and set using the parameters and architecture defined in Tables 3 and 8, plotted in Fig. 10. The results show that the proposed method using model learning can classify cardiac arrhythmias with high accuracy compared to other methods mentioned in this processing field. These results emphasize the efficiency of the proposed algorithm at both scenarios, with and without transfer learning to generate an excellent performance.

4. Conclusion

The application of ECG Arrhythmia Classification is widespread in the healthcare industry, with many potential uses in areas such as clinical diagnosis, disease detection, and patient monitoring. In clinical treatments of cardiac patients, accurate diagnosis of the arrhythmia type is very important and effective in the recovery process. One of the important ways to help in more accurate diagnosis is the use of machine learning techniques, especially a deep learning-based approach. In this paper, a ResNet-34 deep architecture is introduced to classify three types of CVDs using a scalogram of input ECG signals. In the following, the designed ResNet-34 system is combined with the transfer learning technique and the results are considered. Using the proposed architecture can have several prominent effects such as Improved accuracy, transfer learning can help pre-train the deep neural network using a large labeled dataset, which can lead to improved accuracy in ECG classification. In addition, transfer learning can also help the model overcome data sparsity, helping to generalize better to a new dataset. Reduced training time: using pre-trained weights and transfer learning can also reduce the need for training the entire model from scratch,

Table 8
The hyperparameters set for the ResNet34 deep model used transfer learning.

	Epochs	Optimizer	Batch Size	Kernel size	Activation function	Learning rate	Weight decay	#of parameters	# of total layers
ResNet-34	200	Adam	128	3 × 3	Non-Linear	10^{-3}	10^{-4}	0.35 M	34

Table 9
The overall Accuracy the overall accuracy (Acc), Specificity (Spe), Positive prediction rate (Ppr), Sensitivity (Sen), and F-Measure of ECG arrhythmia classification for the RenNet34-based proposed method in combination with transfer learning.

	Acc (%)	Spe (%)	Ppr (%)	Sen (%)	F-Measure (%)
ARR	98.97	100	97.46	97.42	97.44
CHF	98.73	98.62	97.80	98.51	98.15
NSR	98.62	100	98.33	98.76	98.54
Mean	98.77	99.54	98.00	98.23	98.04

Table 10
The confusion matrix for ECG arrhythmia classification of the RenNet34-based proposed method in combination with transfer learning.

ARR (class 1)	**98.97 %**	0.65 %	0.38 %
CHF (class 2)	0.56 %	**98.73 %**	0.71 %
NSR (class 3)	0.89 %	0.49 %	**98.62 %**
	ARR (class 1)	CHF (class 2)	NSR (class 3)

Table 11
The statistical test analysis of the different mentioned method and the RenNet34-based proposed algorithm in combination with transfer learning technique.

		Acc (%)	ρ-value	Mean ρ-value	Mean Acc (%)
ResNet34-based (Proposed)	ARR	**98.97**	0.001	0.003	**98.77**
	CHF	**98.73**	0.005		
	NSR	**98.62**	0.003		

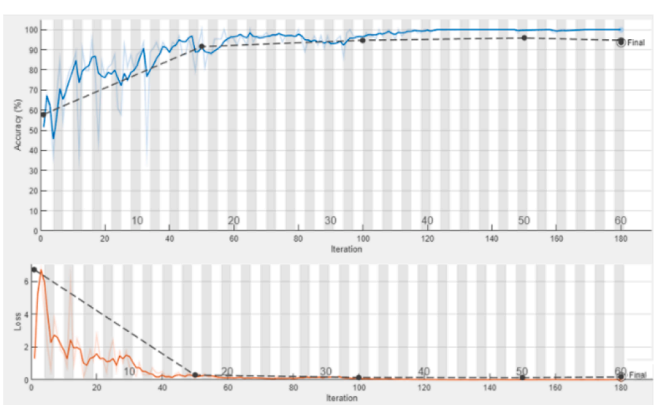

Fig. 10. The proposed ResNet-34 in learning procedure based on transfer learning in the training step.

reducing the training time. This can be especially useful in cases where labeled ECG data is limited or expensive to obtain. Greater generalization ability: by pre-training the model on a large labeled dataset, the model can learn hierarchical and abstract representations of ECG signals that can be generalized to new data. This can help the model identify complex and nonlinear patterns in the data, improving its performance on new ECG signals. The results show that the model presented based on ResNet-34 has better performance in different types of evaluation criteria such as accuracy, and sensitivity, than other detection methods based on different models such as CNN, RNN deep model, and SNMF-based dictionary learning technique.

CRediT authorship contribution statement

Samira Mavaddati: Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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